

# MODEL QUESTION PAPER

REVISION 2010

SUB CODE: 3040

RegNo.....

Signature.....

THIRD SEMESTER DIPLOMA EXAMINATION IN ENGINEERING/TECHNOLOGY

## **Machine Drawing** (Common to ME, AU)

Time : 3 hours

Max Mark.100

Note: 1. All dimensions are in mm

- 2.First angle projection is to be followed
- 3.Missing data if any may be suitably assumed
- 4.Both sides of the drawing sheet may be used.

### **MODULE – 1**

- I) Draw any two views of a hexagonal bolt of size M30X60 with thread length 36mm.  
15 Marks.

OR

- II) Draw two views of single riveted lap joint for 18mm thick plates. Show at least three rivets in plan, indicate all dimensions in terms of the diameter of the rivet. Use snap head rivet.  
15 Marks.

### **MODULE – 2**

- III) Draw the following orthographic views of Socket and Spigot joint shown in figure.1

(a) Top half sectional elevation (b) L.H.S view 30Marks

OR

- IV) Details of a flexible bush type Coupling is shown in figure.2. Draw top half sectional elevation and L H S view.  
30 Marks

### **MODULE -3**

- V) Draw full sectional elevation and top view of the Plummer Block shown in figure. 3  
40 Marks

OR

- VI) Draw the assembled drawing of an IC Engine Connecting rod bottom half sectional Elevation (figure.4)  
40 Marks

## MODULE -4

VII) Drawing the following elementary welding symbols.

- |                           |                       |
|---------------------------|-----------------------|
| 1. Square butt weld       | 2. Single V butt weld |
| 3. Single Bevel butt weld | 4. Single U butt weld |
| 5. Single J butt weld     |                       |

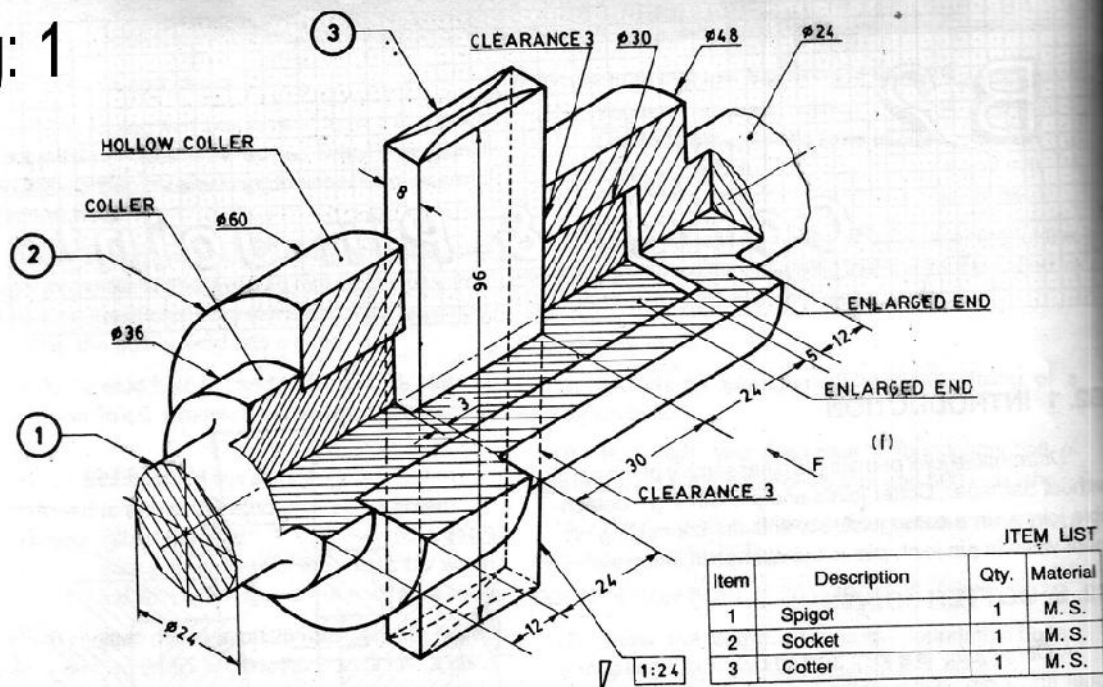
5X3= 15Marks

OR

VIII) Scale Layout of a piping system is shown in figure.5. Prepare the diagrammatic layout of this drawing. All pipes , pipe fittings and valves are 32mm.

15 Marks

**Fig: 1**



**Fig: 2**

Exploded view diagram of a mechanical assembly. The diagram shows the following components and dimensions:

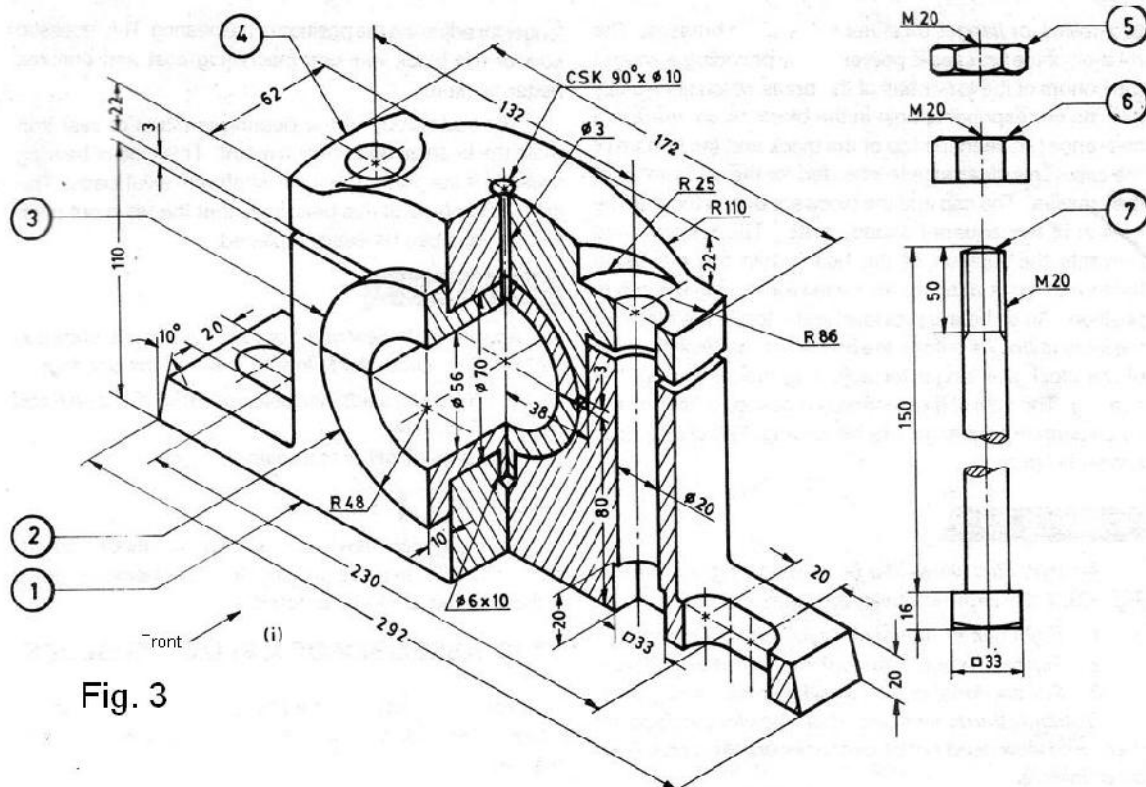
- Component 1:** Shaft end, with a diameter of  $\phi 40$  and a length of 32.
- Component 2:** Pin (Bolt), with a diameter of  $\phi 13$  and a length of 36.
- Component 3:** Key, with a diameter of  $\phi 20$  and a length of 20.
- Component 4:** Flange (Bush side), with a diameter of  $\phi 70$  and a length of 16.
- Component 5:** Bush, with a diameter of  $\phi 13$  and a length of 20.
- Component 6:** Flange (Nut side), with a diameter of  $\phi 70$  and a length of 16.
- Component 7:** Washer, with a diameter of  $\phi 10$  and a thickness of 2.
- Component 8:** Nut, with a diameter of  $\phi 10$  and a length of 16.

Other dimensions and features include:

- SHROUD:** A feature on the shaft end (1) and the flange (6).
- TAPER KEY:** 12 x 8 x 56, shown in two positions.
- R60:** A radius dimension on the shaft end (1) and the flange (6).
- φ160:** A diameter dimension on the shaft end (1).
- φ10:** A diameter dimension on the washer (7).
- φ22:** A diameter dimension on the nut (8).
- M10:** A thread specification on the pin (2) and the nut (8).

| Item | Description          | Qty. | Material |
|------|----------------------|------|----------|
| 1    | Shaft end            | 2    | Steel    |
| 2    | Pin ( Bolt )         | 4    | M. S     |
| 3    | Key                  | 2    | M. S     |
| 4    | Flange ( Bush side ) | 1    | C. I.    |
| 5    | Bush                 | 4    | Rubber   |
| 6    | Flange ( Nut side )  | 1    | C. I.    |
| 7    | Washer               | 4    | M. S     |
| 8    | Nut                  | 4    | M. S     |

| Item | Description          | Qty. | Material |
|------|----------------------|------|----------|
| 1    | Shaft end            | 2    | Steel    |
| 2    | Pin ( Bolt )         | 4    | M. S.    |
| 3    | Key                  | 2    | M. S.    |
| 4    | Flange ( Bush side ) | 1    | C. I.    |
| 5    | Bush                 | 4    | Rubber   |
| 6    | Flange ( Nut side )  | 1    | C. I.    |
| 7    | Washer               | 4    | M. S.    |
| 8    | Nut                  | 4    | M. S.    |



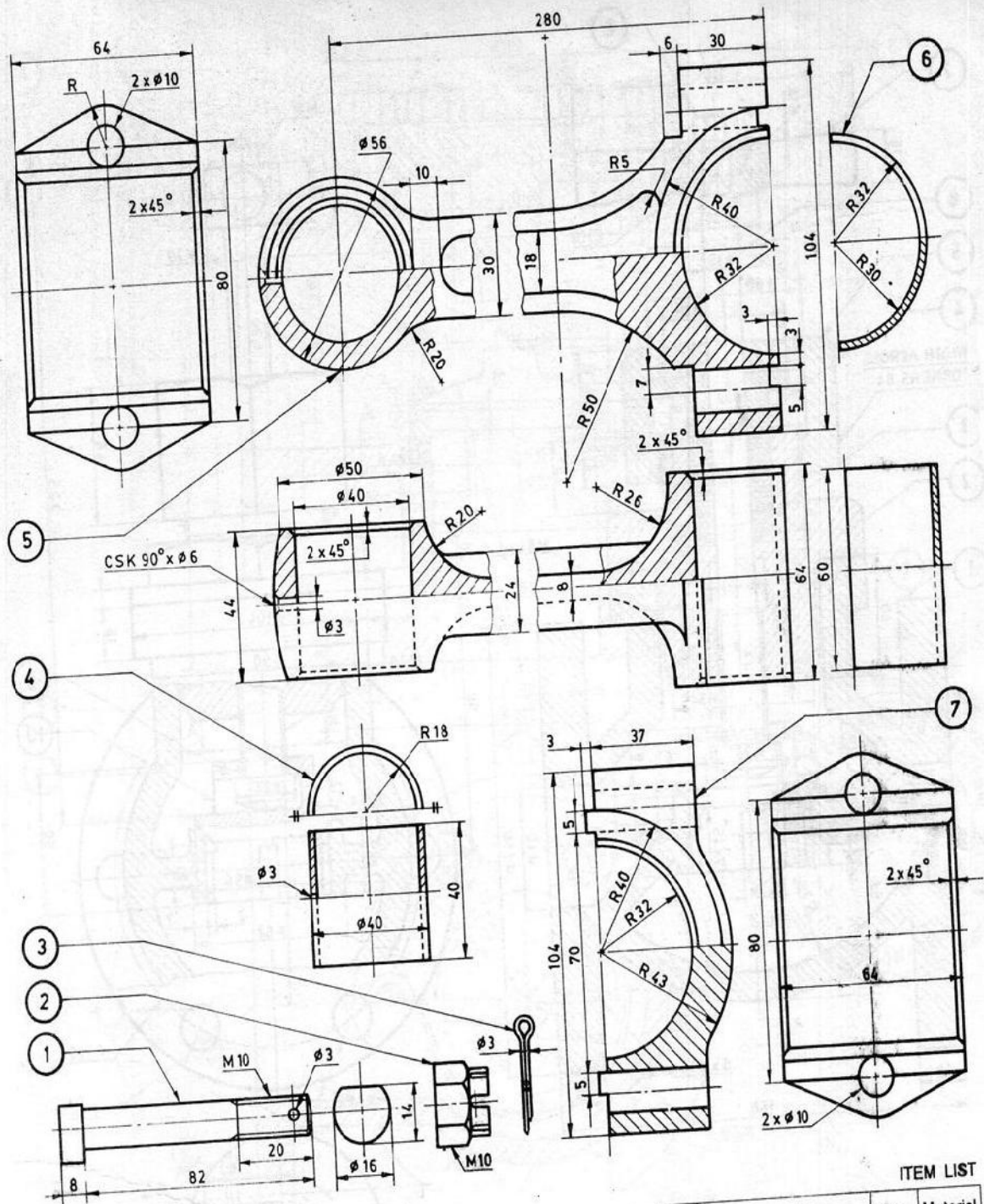


Fig.4

ITEM LIST

| Item | Description      | Qty. | Material |
|------|------------------|------|----------|
| 1    | Bolt             | 2    | M. S.    |
| 2    | Lock nut         | 2    | M. S.    |
| 3    | Split pin        | 2    | M. S.    |
| 4    | Bush (Small end) | 1    | Bronze   |
| 5    | Body             | 1    | Steel    |
| 6    | Bush (Big end)   | 2    | Bronze   |
| 7    | Cap              | 1    | Steel    |

Fig.5

